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Problem Solving Protocol

Revealing hidden regulatory dependencies: multi-perspective graph learning for single-cell gene regulatory network inference

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Abstract

Gene regulatory networks (GRNs) inform analyses of cellular state transitions, regulatory mechanisms, and disease processes. With the rapid development of single-cell sequencing technologies, accurate inference of GRNs from complex and high-dimensional single-cell transcriptomic data remains a core challenge. However, the effective use of multi-level structural and expression features among genes remains a major obstacle to improving inference accuracy. This study presents ATFGRN, an adaptive topology-feature fusion graph neural framework that integrates features from three complementary perspectives for accurate prediction of gene regulatory relationships. The subgraph structure encoding module focuses on local subgraphs of regulatory relationships and identifies structural patterns and topological dependencies. The expression-guided module integrates the gene expression matrix with the original regulatory network and employs a graph convolutional network with a self-attention mechanism to examine interactions between expression dynamics and network topology. The similarity structure module derives similarity information between genes through a KNN graph combined with a graph attention mechanism, which helps detect regulatory pairs with similar expression patterns that lack explicit structural links. Features from these three branches are fused through an attention-based weighting mechanism. This fusion achieves complementary integration of structural, expression, and similarity perspectives and produces more informative regulatory features for prediction. Evaluations on single-cell transcriptomic datasets across four types of networks show that ATFGRN improves AUROC performance by 5.09% over existing approaches, which confirms the effectiveness and applicability of its multi-perspective fusion strategy in GRN inference tasks.

Keywords: gene regulation network; graph learning; multi-perspective fusion; ScRNA-seq

Introduction

Gene regulatory networks (GRNs) describe regulatory interactions among genes, with transcription factor (TF)–mediated regulation constituting an important subset. These networks serve as a core framework for understanding cellular development, differentiation, and disease mechanisms [1]. Identifying and analyzing these regulatory interactions help reveal how TFs influence gene expression under different conditions, with implications for precision medicine, cancer research, and systems biology [2]. The rapid advancement of single-cell RNA sequencing (scRNA-seq) technology offers new opportunities for GRN reconstruction. Compared with traditional bulk sequencing data, single-cell data provide higher resolution for characterizing cellular heterogeneity, revealing subtle differences between cell states, and offering a more precise foundation for modeling gene regulatory relationships [3, 4].

Despite considerable progress, GRN inference from scRNA-seq data remains challenging because dropout events introduce sparsity and noise [5]. Traditional approaches such as GENIE3 [6], GRNBoost2 [7], and SCENIC [8] improve interpretability and efficiency but are vulnerable to expression noise and sparsity. Meanwhile, differential equation-based methods like SCODE [9] depend heavily on pseudotime estimation, which can be difficult to infer accurately. With the rapid advancement of deep learning techniques [10], several methods such as CNNC [11] and DGRNs [12] learn nonlinear dependencies from high-dimensional data and achieve robust performance. However, these methods primarily focus on expression information while overlooking the underlying regulatory topology. To overcome this limitation, graph neural networks (GNNs) have been incorporated into GRN inference [13]. GNN-based frameworks including GENELink [14], GNE [15], GATCL [16], and scMGATGRN [17] exploit graph attention and multi-view mechanisms to integrate structural and expression-level features.

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These approaches strengthen the representation of regulatory dependencies among genes and significantly improve inference accuracy across diverse single-cell datasets. Although GNN-based methods have shown strong potential in GRN reconstruction, several limitations persist. Many models combine graph structure with expression data through shallow fusion strategies, which restrict their ability to explore regulatory information from multiple perspectives. Some approaches depend heavily on prior regulatory networks, leading to poor generalization across cell types. Moreover, as GNN depth increases, over-smoothing occurs, making node representations indistinguishable and reducing their discriminative power [18–20]. Therefore, there is a need for a new GRN inference framework that integrates multi-perspective regulatory information while maintaining a balance between local and global structures.

To address these issues, we propose a multi-perspective fusion framework for GRN inference named ATFGRN. The framework integrates diverse expression and structural information to improve the accuracy and robustness of GRN prediction. It extracts features for each candidate TF-Target gene pair from three complementary perspectives. The first perspective constructs a second-order subgraph for each candidate edge to obtain local topological information. The subgraph is structurally labeled and encoded through a graph convolutional encoder to generate structural features. The second perspective combines the normalized single-cell expression matrix with the original regulatory graph and applies a graph convolutional network with self-attention to encode node contexts and derive expression-driven regulatory representations. The third perspective builds a KNN similarity graph based on the expression matrix, initializes embeddings with the Node2Vec algorithm [21], and passes them through a two-layer GAT followed by an LSTM module to detect potential regulatory links between genes with similar expression profiles. After the three branches extract their respective features, ATFGRN employs a shared-query attention mechanism to learn the importance of each perspective and fuse the features into a unified representation. Finally, a multilayer perceptron (MLP) maps the fused vector into a scalar score for regulatory prediction. Through the integration of structure, expression, and similarity information, ATFGRN enhances both interpretability and predictive accuracy in single-cell GRN inference.

Materials and methods

Datasets and preprocessing of raw data

To comprehensively evaluate the performance of ATFGRN, we employed seven single-cell sequencing datasets [22], including two from humans and five from mice, specifically: (i) human embryonic stem cells; (ii) human mature hepatocytes; (iii) mouse dendritic cells; (iv) mouse embryonic stem cells; (v) erythroid lineage mouse hematopoietic stem cells; (vi) granulocyte-monocyte lineage hematopoietic stem cells in mice; and (vii) lymphoid lineage hematopoietic stem cells in mice. Each dataset contains three types of gene regulatory networks: the STRING network [23], constructed based on protein functional interactions, serving as a proxy for potential functional associations between genes; cell type-specific ChIP-seq networks [24, 25], derived from ChIP-seq experiments matching the cell type of the target dataset, indicating the physical binding potential of TFs; and non-specific ChIP-seq networks [26–28], built by integrating ChIP-seq data from multiple cell types or tissues, encompassing a broader range of putative regulatory interactions. Additionally, the mESC dataset includes a network based on loss-of-function and gain-of-function (LOF/GOF) experiments, providing experimental evidence of the functional consequences of TF perturbations [29]. Although these networks do not represent perfect ground truths, they collectively form a widely used and accepted benchmark suite for fair and consistent evaluation of GRN inference methods.

Following the preprocessing method provided by BEELINE [22], the seven datasets mentioned above were processed accordingly. First, genes expressed in less than 10% of cells were removed, and genes with Bonferroni-corrected p-values of expression variance below 0.01 were considered significantly variable. Next, all significantly variable TFs were selected. Non-TF genes passing the p-value filter were then ranked in descending order based on their expression variance, from which the top 500 and top 1000 significantly variable genes were selected separately. These were combined with the significantly variable TFs to form the TFs + 500 and TFs + 1000 datasets. Detailed distributions of each dataset are provided in Table 1. Different network types exhibit distinct densities and structural characteristics, and corresponding sampling and partitioning strategies are applied during dataset construction. The detailed procedures for data grouping and negative sampling are described in the Supplementary file.

The ATFGRN framework

In this study, the GRN inference task based on single-cell transcriptomic data is formulated as a link prediction problem [30]. Each gene serves as a node in the graph, with node features derived from scRNA-seq profiles, while potential regulatory relationships between TFs and target genes correspond to edges. The aim of GRN inference is to identify true regulatory edges from all candidate TF-Target pairs. As illustrated in Fig. 1, ATFGRN is a supervised deep learning framework that integrates scRNA-seq gene expression data and a prior GRN to generate informative edge representations (see Fig. 1a). The framework extracts gene expression features and structural information from multiple perspectives to improve the recognition of regulatory relationships. ATFGRN is organized into three feature extraction modules and one weighted fusion module. The first branch, following subgraph-based strategies [31–34], constructs a local subgraph for each candidate TF-Target pair and employs Double-Radius Node Labeling (DRNL) with GCNs to derive structural semantics (see Fig. 1b). The second branch encodes features from the prior GRN adjacency matrix and expression data through a TransformerConv module with a self-attention mechanism, which emphasizes genes with stronger regulatory influence and enhances edge prediction accuracy (see Fig. 1c). The third branch generates a first-order similarity graph by calculating Euclidean distances among gene expression profiles and applies Node2Vec and GAT to obtain embeddings that represent shared expression characteristics (see Fig. 1d). Edge representations are obtained by aggregating features of TF-Target pairs, while the fusion module assigns adaptive weights to combine outputs from the three branches (see Fig. 1e). This design enables ATFGRN to reveal both local and global cues for GRN reconstruction and improve predictive robustness.

Local topology subgraph encoding module

To extract local structural features reflecting regulatory dependencies, we construct local subgraphs for candidate edges, and encode their topology using structural labeling and graph convolutional networks. For each candidate edge's local subgraph extraction, we start from the original gene regulatory network adjacency matrix $A \in R^{N\times N}$ and extract its second-order neighborhood subgraph in the graph. This choice provides a balanced depth that captures indirect regulatory contexts while avoiding

Table 1. The statistics of four ground-truth networks with TFs and 500 (1000) most-varying genes

Cell types	Cells	STRING				Non-Specific			
		TFs	Genes	Edges	Density	TFs	Genes	Edges	Density
hESC	758	343(351)	511(695)	4257(5149)	0.024(0.021)	283(292)	753(1138)	3441(4617)	0.016(0.014)
hHEP	425	409(414)	646(874)	7523(9003)	0.028(0.024)	322(332)	825(1217)	4129(5351)	0.015(0.013)
mDC	383	264(273)	479(664)	4815(5898)	0.038(0.032)	250(254)	634(969)	3067(3918)	0.019(0.016)
mESC	421	495(499)	638(785)	7762(8479)	0.024(0.021)	516(522)	890(1214)	6893(8030)	0.015(0.013)
mHSC-E	1071	156(161)	291(413)	1371(1826)	0.029(0.027)	144(147)	442(674)	1425(1960)	0.022(0.020)
mHSC-GM	889	92(100)	201(344)	748(1311)	0.040(0.037)	82(88)	297(526)	743(1358)	0.030(0.029)
mHSC-L	847	39(40)	70(81)	137(154)	0.048(0.045)	35(37)	164(192)	279(317)	0.048(0.043)
Cell types	**Cells**	**Specific**				**LOF/GOF**			
		TFs	**Genes**	**Edges**	**Density**	**TFs**	**Genes**	**Edges**	**Density**
hESC	758	34(34)	815(1260)	4545(7084)	0.164(0.165)	-	-	-	-
hHEP	425	30(31)	874(1331)	9939(15558)	0.379(0.377)	-	-	-	-
mDC	383	20(21)	443(684)	756(1193)	0.085(0.082)	-	-	-	-
mESC	421	88(89)	977(1385)	29,613(42795)	0.345(0.347)	34 (34)	774 (1098)	4169(5742)	0.158(0.154)
mHSC-E	1071	29(33)	691(1177)	11,557(21975)	0.578(0.566)	-	-	-	-
mHSC-GM	889	22(23)	618(1089)	7364(14135)	0.543(0.565)	-	-	-	-
mHSC-L	847	16(16)	525(640)	4398(5180)	0.525(0.507)	-	-	-	-

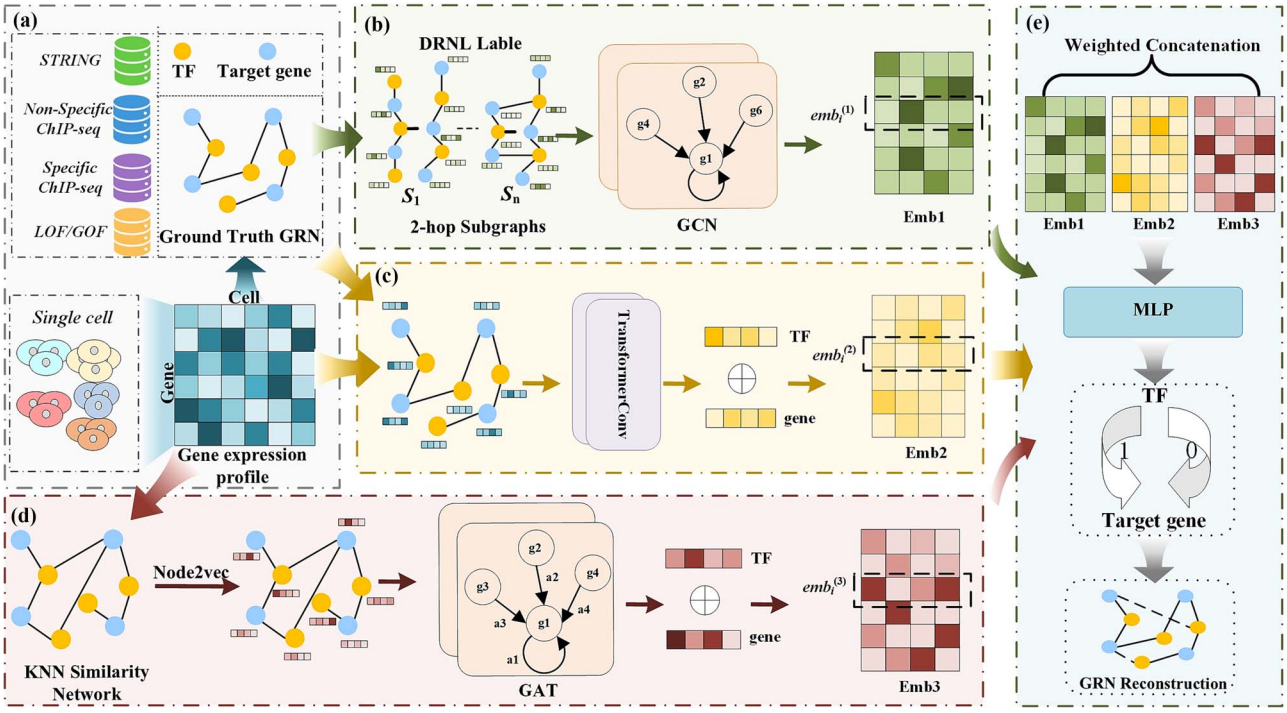


Figure 1. The overall architecture of ATFGRN. (a) Preprocessing module takes the gene-cell expression matrix and ground-truth GRN as input. (b) Local topology subgraph encoding module constructs local subgraphs and applies DRNL and GCN layers to extract structural features. (c) Expression-guided attention module encodes expression and adjacency information through TransformerConv with self-attention. (d) Expression similarity embedding module builds a similarity graph from expression distances and learns embeddings with Node2Vec and GAT. (e) Feature fusion and link prediction module adaptively combines the outputs of the three branches to reconstruct regulatory relationships.

the noise and computational overhead introduced by larger receptive fields. The resulting subgraph includes the candidate TF and gene nodes, as well as nodes and edges that are directly or indirectly connected within two hops, thereby preserving local structural information that reflects potential regulatory relationships.

To effectively distinguish the structural roles of nodes within the local subgraph relative to the candidate edge, we introduce the DRNL strategy. This strategy assigns an integer label to each node in the subgraph through a structural labeling function $f_l : V \to N$, encoding its relative topological position in the graph. The specific formula is as follows:

$$f_l(i) = 1 + \min(d_x, d_y) + \left(\frac{d}{2}\right)\left[\left(\frac{d}{2}\right) + (d\%2) - 1\right] \quad (1)$$

where $d_x = d(i, x)$ and $d_y = d(i, y)$ denote the shortest path distances from node i in the subgraph to the two endpoints x and y

of the candidate edge, respectively, and $d = d_x + d_y$ represents their sum. DRNL assigns the unique label 1 to the two central nodes at the ends of the candidate edge, namely the TF and the target gene, to highlight their key roles in the regulatory relationship. Other neighboring nodes are assigned distinct integer labels based on their combined distances to these two central nodes. This structural labeling explicitly differentiates the roles and relative positions of nodes within the local graph, providing more discriminative topological information to support gene regulatory relationship prediction.

Next, we encode the structural labels of each node in the subgraph as one-hot vectors to serve as node features X. The dimensionality of these one-hot vectors corresponds to the total number of distinct DRNL labels present across all extracted subgraphs, ensuring that each structural role is uniquely represented. We then employ a graph convolutional network (GCN) encoder to extract deeper structural features from each subgraph. Let $G = \{V, E\}$ denote the prior GRN, where V represents the set of nodes in the subgraph and E represents the set of edges indicating regulatory relationships between gene nodes. In this step, the adjacency matrix refers specifically to the extracted local subgraph. The graph convolution operation aggregates the features of each node's neighbors within the subgraph to learn semantic representations in the topological context [35]. The graph convolution is defined as follows:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \tag{2}$$

where $\tilde{A} = A + I_N$ represents the adjacency matrix with added self-loops, with I_N denoting the identity matrix of size N, allowing the node to retain its original features, and $\tilde{D}$ is the corresponding degree matrix. $H^{(l)} \in \mathbb{R}^{N\times F}$ denotes the node embeddings at the l-th layer, where $H^{(0)} = X$. $W^{(l)}$ is the learnable weight matrix at layer l. The function $\sigma(\cdot)$ is a nonlinear activation function, for which Relu is used in this work.

To convert variable-sized subgraphs into fixed-dimensional vector representations, we first compute the $L2$ norm of each node's embedding obtained after multiple layers of graph convolution within each subgraph. This norm serves as an importance score of the node relative to the candidate edge. Nodes in the subgraph are then ranked based on their $L2$ norms, and the top k nodes are selected as representative structures of the subgraph. This process is formally described by the following equation:

$$P_i = Top_k\left(\{\| e_j\|_2 \mid j \in \mathcal{V}_i\}\right) \tag{3}$$

where P_i denotes the set of the top k nodes selected from the i-th subgraph, $\| e_j\|_2$ represents the $L2$ norm of the embedding vector of node j, and $\mathcal{V}_i$ is the set of all nodes in the i-th subgraph. To ensure a consistent dimensionality of subgraph representations, the value of k is set to the number of nodes in the k-th largest subgraph (by node count) in the training set, thereby adaptively adjusting the subgraph size.

Finally, each subgraph is flattened into a one-dimensional vector $z_i \in \mathbb{R}^{k\cdot d}$, where d is the dimension of the node embeddings. We then use a two-layer perceptron to map the subgraph representation into the final embedding space. The process is described as follows:

$$emb_i^{(1)} = W_2 \cdot ReLU\left(W_1 \cdot z_i + b_1\right) + b_2 \tag{4}$$

Here, $emb_i^{(1)}$ represents the final structural feature of the subgraph corresponding to the i-th candidate edge, capturing the latent regulatory cues between the TF and the target gene within the local structure.

Expression-guided attention module

In this module, our goal is to leverage gene expression features together with the known GRN to uncover potential regulatory relationships from the perspective of gene expression. To obtain initial expression features for each gene node, we normalize the acquired ScRNA-seq data, resulting in a gene expression matrix. To fully capture regulatory context information within the graph structure, we introduce a graph convolution method with an attention mechanism-TransformerConv [36] to extract features for each node. This method dynamically adjusts the importance of neighboring nodes to the target node based on multi-head dot-product attention, effectively capturing dependencies among nodes under the known structure. The specific update formula is as follows:

$$x_i' = W_1 x_i + \sum_{j\in\mathcal{N}(i)} \alpha_{i,j} W_2 x_j \tag{5}$$

where x_i denotes the gene expression feature of node i, $\mathcal{N}(i)$ is the set of neighboring nodes of node i, and the attention weight $\alpha_{i,j}$ is computed via dot-product attention, as defined below:

$$\alpha_{i,j} = softmax\left(\frac{(W_3 x_i)^\top (W_4 x_j)}{\sqrt{d}}\right) \tag{6}$$

where W_3 and W_4 are the linear transformation matrices for the query and key, respectively, x_j is the input feature of the neighboring node j, and d is the scaling factor.

To enable the model to learn gene embeddings from multiple perspectives, we employ a multi-head attention mechanism to obtain the final gene representations. Each head independently learns a set of attention coefficients and generates a corresponding set of feature representations. The final node representation is computed as the average of the outputs from all attention heads. The formulation is as follows:

$$h_i = \frac{1}{K}\sum_{k=1}^{K} x_i'^{(k)} \tag{7}$$

where K denotes the number of attention heads, and h_i represents the embedding of node i after applying multi-head attention.

Finally, we obtain the edge feature representation by summing the feature vectors of the node pair (TF, Target) corresponding to the candidate edge. Specifically, for the i-th candidate edge, which consists of a TF node and a target gene (Target) node, their respective feature representations are denoted as h_{TF_i} and h_{Target_i}. The edge feature is computed as follows:

$$emb_i^{(2)} = h_{TF_i} + h_{Target_i} \tag{8}$$

where $emb_i^{(2)}$ denotes the final representation of the i-th candidate edge in this module. This feature representation is based on the known GRN structure and emphasizes the role of gene expression features in regulatory relationships, thereby enhancing the model's understanding of the existing regulatory map and improving prediction accuracy.

Expression similarity embedding module

Prior studies confirm that a high correlation in expression levels between a TF and its target gene often signals a possible regulatory link across various samples [37]. Based on this insight, we investigate gene–gene regulatory relationships by comparing expression similarity. Before computing similarity, the scRNA-seq expression matrix is standardized using Z-score normalization to ensure comparable feature scales. This process depends solely on normalized scRNA-seq data and avoids reliance on predefined GRNs.

We start by computing the Euclidean distance between gene expression vectors. For each gene, we identify its K nearest neighbors with the smallest distances. These top-K neighbors form the first-order adjacency edges, constructing a graph structure grounded in expression similarity $A_{knn} \in \mathbb{R}^{n\times n}$, here n is the total number of genes. To derive informative node embeddings, we employ the Node2Vec algorithm [21], which generates a 32-dimensional embedding for each gene, capturing its structural and contextual properties within the similarity graph.

After generating these initial representations, we apply a two-layer Graph Attention Network (GAT) for further refinement. GAT dynamically allocates weights to neighbor nodes through an attention mechanism. This adaptive weighting highlights key expression patterns from neighbors [38]. In each GAT layer, we calculate the attention coefficients between a gene node and its adjacent nodes as follows:

$$\alpha_{ij} = \frac{\exp\left(LeakyReLU\left(\vec{a}^{T}\left[W\vec{h}_i \parallel W\vec{h}_j\right]\right)\right)}{\sum_{k\in\mathcal{N}_i}\exp\left(LeakyReLU\left(\vec{a}^{T}\left[W\vec{h}_i \parallel W\vec{h}_k\right]\right)\right)} \tag{9}$$

where LeakyReLU is the activation function with a negative slope of 0.2, $\mathcal{N}_i$ denotes the set of first-order neighbors of gene i, W is initialized using Xavier initialization, $a \in \mathbb{R}^{2F'}$ is a learnable weight vector, F' presents the dimensionality of gene features, and *exp* denotes the exponential function.

We then perform feature aggregation by combining the representations of neighboring nodes:

$$\vec{h}'_i = \sigma\left(\sum_{j\in\mathcal{N}_i}\alpha_{ij}W\vec{h}_j\right) \tag{10}$$

To better retain diverse semantic signals across GAT layers, we introduce a skip connection mechanism. Instead of simple averaging, we feed the outputs from two GAT layers into a unidirectional LSTM. This LSTM learns the hierarchical importance of each layer's features [39]. As a result, the final embedding integrates heterogeneous structural cues captured at different receptive fields more effectively.

To represent candidate edges, we follow the same procedure as the previous module. For each TF-Target gene pair, we sum their final node embeddings:

$$emb_i^{(3)} = z_i^{TF} + z_i^{Target} \tag{11}$$

where z_i^{TF} and z_i^{Target} denote the final embeddings of the TF and Target nodes corresponding to the i-th candidate edge, respectively. The resulting vector $emb_i^{(3)}$ serves as the expression-similarity-based edge feature.

Feature fusion and link prediction

To integrate the candidate edge features derived from three distinct perspectives, an attention-based fusion mechanism is employed to form the final edge representation for gene regulatory relationship prediction. The attention coefficient is computed as follows:

$$e_i^{(j)} = q^{\top}\cdot\tanh\left(W_j\cdot emb_i^{(j)} + b_j\right), j\in\{1,2,3\} \tag{12}$$

where $q \in \mathbb{R}^{1\times F'}$ denotes the shared linear transformation layer across the three branches, $emb_i^{(j)}$ represents the final embedding of candidate edge i in the j-th branch, and tanh is the activation function. W_j and b_j are the feature mapping matrix and bias term of the j-th branch, respectively. $e_i^{(j)}$ denotes the attention coefficient of candidate edge i for branch j.

The softmax function normalizes all attention coefficients from the three branches to produce fusion weights:

$$\alpha_i^{(j)} = \frac{\exp\left(e_i^{(j)}\right)}{\sum_{k=1}^{3}\exp\left(e_i^{(k)}\right)}, j\in\{1,2,3\} \tag{13}$$

After obtaining the normalized weights, the framework aggregates the candidate edge features from each perspective through weighted concatenation. This operation yields the final embedding vector of the edge:

$$emb_i = \alpha_i^{(1)}\cdot emb_i^{(1)} \parallel \alpha_i^{(2)}\cdot emb_i^{(2)} \parallel \alpha_i^{(3)}\cdot emb_i^{(3)} \tag{14}$$

where $\parallel$ denotes vector concatenation, and emb_i is the fused feature representation of the i-th candidate edge obtained from the three branches.

The fused edge features are subsequently passed through a multilayer perceptron (MLP), which outputs the probability that a regulatory interaction exists between a TF and a target gene. The overall training objective integrates four loss components: one derived from the fused features and three from the individual branches. The total loss is expressed as:

Each term employs a binary cross-entropy loss function,

$$\mathcal{L}_{total} = \mathcal{L}_{fusion} + \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 \tag{15}$$

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i\log\left(\hat{y}_i\right) + \left(1-y_i\right)\log\left(1-\hat{y}_i\right)\right] \tag{16}$$

where N is the number of genes, y_i is the true label of the i-th candidate edge, and $\hat{y}_i$ refers to its predicted probability derived from the sigmoid activation. This formulation ensures that both global and branch-specific representations contribute effectively to the final prediction accuracy.

Evaluation strategies

In this study, we employed two metrics to evaluate model performance: the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC). AUROC measures the overall discrimination ability of the model between positive and negative samples across different classification thresholds, while AUPRC places more emphasis on the identification of positive samples under imbalanced data conditions. Thus, AUPRC holds greater reference value in GRN inference tasks where the positive-to-negative sample ratio is

severely imbalanced. Additionally, metrics such as true positive rate (TPR), false positive rate (FPR), precision, and recall are used, defined as follows:

$$TPR = \frac{TP}{TP + FN}$$
$$FPR = \frac{FP}{FP + TN}$$
$$Pre = \frac{TP}{TP + FP}$$
$$Recall = \frac{TP}{TP + FN} \quad (17)$$

Here, TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively.

Results

Performance on benchmark datasets

To evaluate the performance of ATFGRN, we conduct comparative experiments on seven single-cell benchmark datasets with four types of ground-truth regulatory networks. The comparison covers seven representative approaches that represent both traditional and deep learning-based models. PCC [40] measures co-expression similarity through Pearson correlation coefficients; GENIE3 [6] employs random forests to infer TF-Target relationships based on feature importance; SCODE [9] models dynamic expression regulation via linear ordinary differential equations; DeepSEM [41] integrates structural equation modeling into β-VAE for joint representation learning and regulatory inference; GENELink [14] applies graph attention to emphasize key nonlinear dependencies; GATCL [16] combines graph convolution and multi-head attention for contextual expression representation; and scMGATGRN [17] adaptively fuses multi-view embeddings through view-level attention to quantify gene–gene interactions.

As shown in Fig. 2, within the first group of networks (STRING, Non-Specific ChIP-seq, and LOF/GOF), ATFGRN demonstrates a distinct advantage through its deep integration of graph structure and gene expression information. On the 15 datasets with TFs + 500, it achieves an average AUROC improvement of 5.09% over the second-best method, scMGATGRN. This advantage remains clear in the TFs + 1000 datasets, where ATFGRN attains an average AUROC exceeding 0.88, more than 5% higher than all other methods. Across all 30 datasets, ATFGRN achieves the highest AUROC scores, indicating strong generalization ability and robustness. Regarding AUPRC (Fig. 2b), ATFGRN also performs outstandingly, surpassing all comparison methods on 27 out of 30 datasets. On the STRING network, which features dense connections and severe class imbalance, ATFGRN benefits from subgraph structural representation, yielding an average AUPRC of 0.53, much higher than GATCL's 0.32. These results verify its stability and precision in complex regulatory conditions. In the second group of networks (Specific), ATFGRN maintains strong performance. Compared with scMGATGRN, it achieves equal or better AUROC scores on 78.6% (11/14) of the datasets, with an average AUPRC improvement of 2.67%. All models tend to reach higher AUPRC scores on Specific networks than on the first group, likely due to better data quality and lower noise levels that facilitate learning of regulatory patterns.

To further analyze the performance variation across datasets, Fig. 3 illustrates the AUROC distributions of the eight models on the three network types (STRING, Non-Specific, and Specific). The violin plots depict the distribution of AUROC scores across the seven single-cell datasets for each method. Graph-based learning methods, including ATFGRN, GATCL, scMGATGRN, and GENELink, achieve consistently higher AUROC values than traditional algorithms such as PCC, GENIE3, and SCODE. This difference highlights the advantage of graph neural networks in representing complex regulatory dependencies among genes.

Through the joint design integrating structure, expression, and similarity perspectives, ATFGRN consistently achieves strong results in GRN inference tasks. It not only leads in AUROC but also demonstrates improved adaptability to imbalanced data in AUPRC, enhancing the identification precision of key regulatory relationships. These findings confirm the effectiveness and reliability of ATFGRN for high-resolution GRN inference in single-cell analysis.

Ablation experiment

To evaluate the contribution of each branch module to the overall performance of ATFGRN, three ablation variants are designed: (i) ATFGRN w/o K, (ii) ATFGRN w/o S, and (iii) ATFGRN w/o G. In these variants, w/o K denotes the removal of the KNN-based similarity network to assess the effect of gene expression similarity on regulatory prediction. w/o S refers to excluding the subgraph structure-aware module, which examines the role of local topological information in describing regulatory dependencies. w/o G removes the TransformerConv-based global feature extraction branch, which evaluates the contribution of semantic information derived from the original GRN and expression data to prediction accuracy.

These three variants are trained on seven benchmark scRNA-seq datasets constructed with the STRING network and compared with the full ATFGRN under identical hyperparameter settings. Figure 4 shows that removing any branch results in a clear performance decline. Among them, removing the subgraph branch (w/o S) causes the most notable degradation, with an average AUROC reduction of 4.27%, underscoring its key role in representing local structural and higher-order relational patterns. The removal of the TransformerConv branch (w/o G) produces an average AUROC drop of 1.61%, indicating that global semantic information from the regulatory network and expression data remains vital for accurate prediction. By contrast, eliminating the KNN similarity branch (w/o K) yields a 1.74% decrease, confirming that this module complements graph structural cues and refines node representations. Detailed numerical results are listed in Supplementary Table S1.

Overall, the three feature extraction branches in ATFGRN cooperate effectively and contribute distinct yet complementary information, thereby enhancing generalization capability and improving predictive accuracy across diverse datasets.

Hyperparameter tuning

The performance of the proposed model depends on several key hyperparameters, including the number of attention heads in TransformerConv, the number of GNN layers, and the learning rate. Hyperparameter tuning is conducted on seven benchmark cell-type-specific datasets, where AUROC and AUPRC serve as evaluation metrics. Detailed numerical outcomes are provided in Supplementary Tables S2–S4.

The influence of these three parameters is analyzed to determine their optimal settings. For the attention heads in TransformerConv, values in {1, 4, 8, 12, 16} are examined. As shown in Fig. 5a, the model achieves the highest AUROC and AUPRC values on most datasets when the number of heads equals 8, indicating

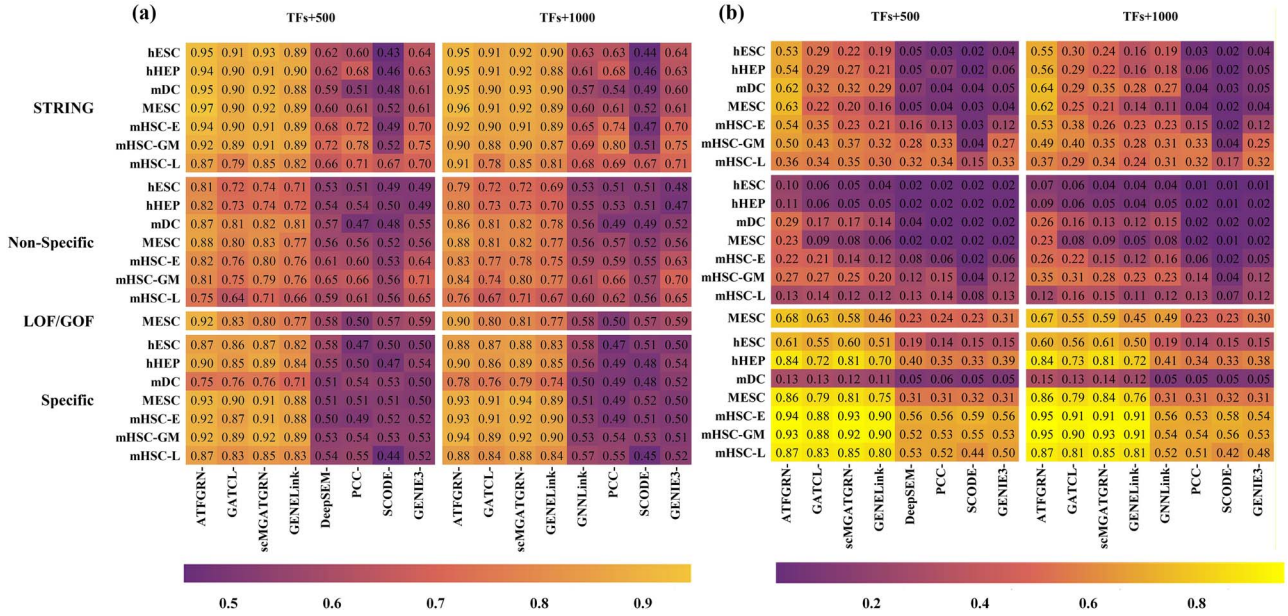


Figure 2. AUROC (a) and AUPRC (b) results for the performance comparison of all models on all benchmark datasets.

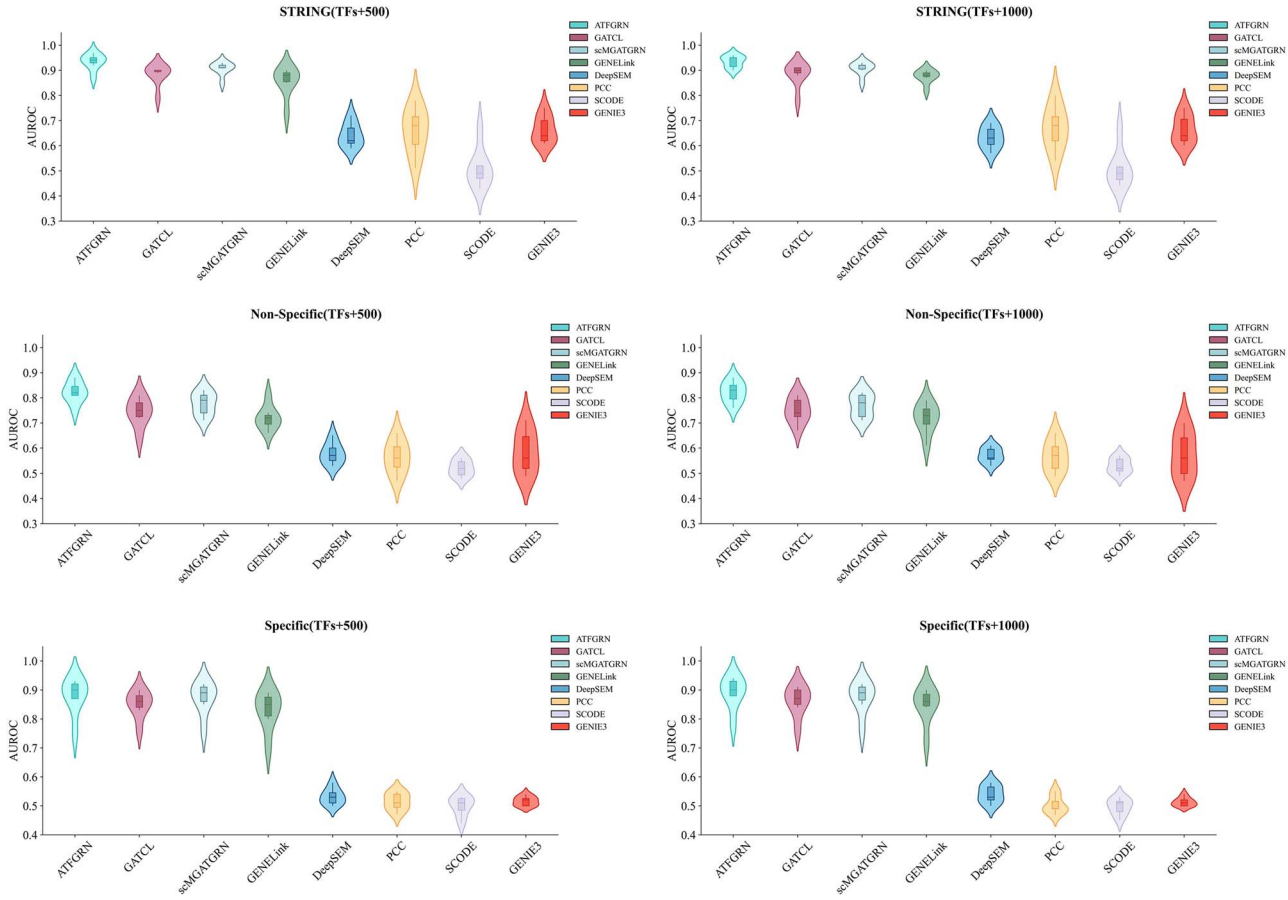


Figure 3. Distribution of AUROC scores on single-cell datasets illustrated by violin plots.

that a moderate number of heads effectively integrates information from multiple perspectives. Regarding the GNN layers, Fig. 5b shows that performance peaks at two layers, suggesting that this depth balances feature propagation and local structure representation. Adding more layers leads to over-smoothing, which reduces discrimination among node embeddings. For the learning rate, Fig. 5c demonstrates that 0.001 yields the best and most stable results. Smaller rates slow convergence, whereas larger ones cause optimization instability. Overall, these findings confirm that appropriate parameter choices-eight attention heads, two GNN layers, and a learning rate of 0.001-produce stable and accurate performance across single-cell datasets.

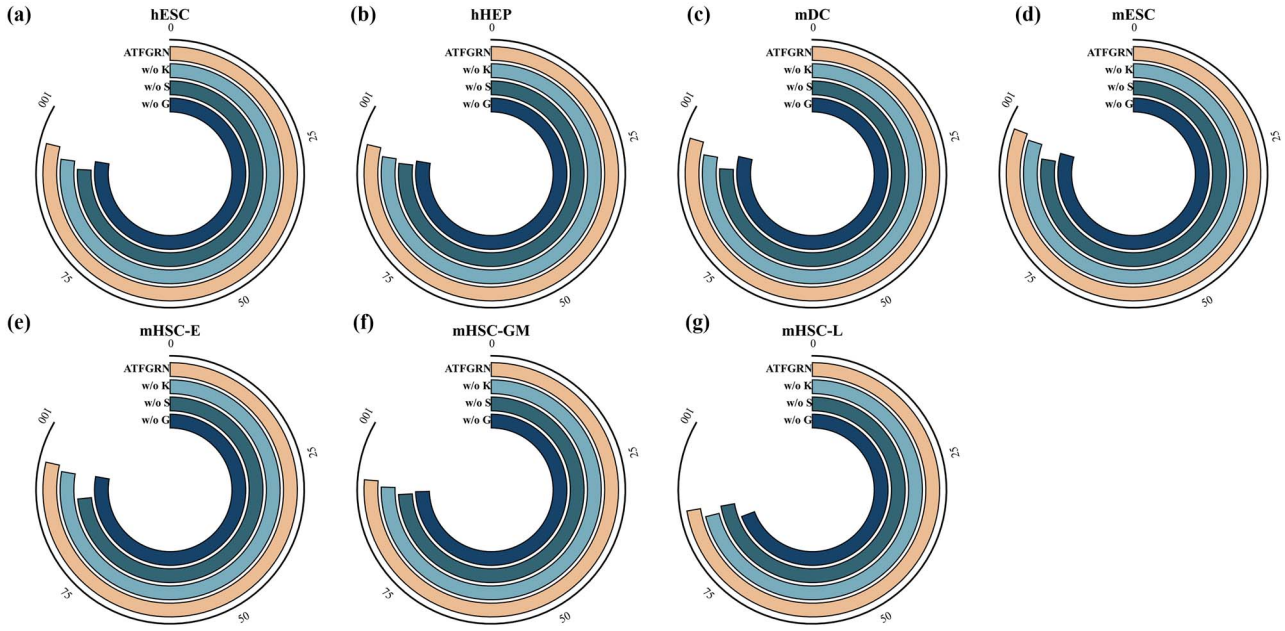


Figure 4. AUROC results of ATFGRN variants in ablation experiments on multiple single-cell datasets.

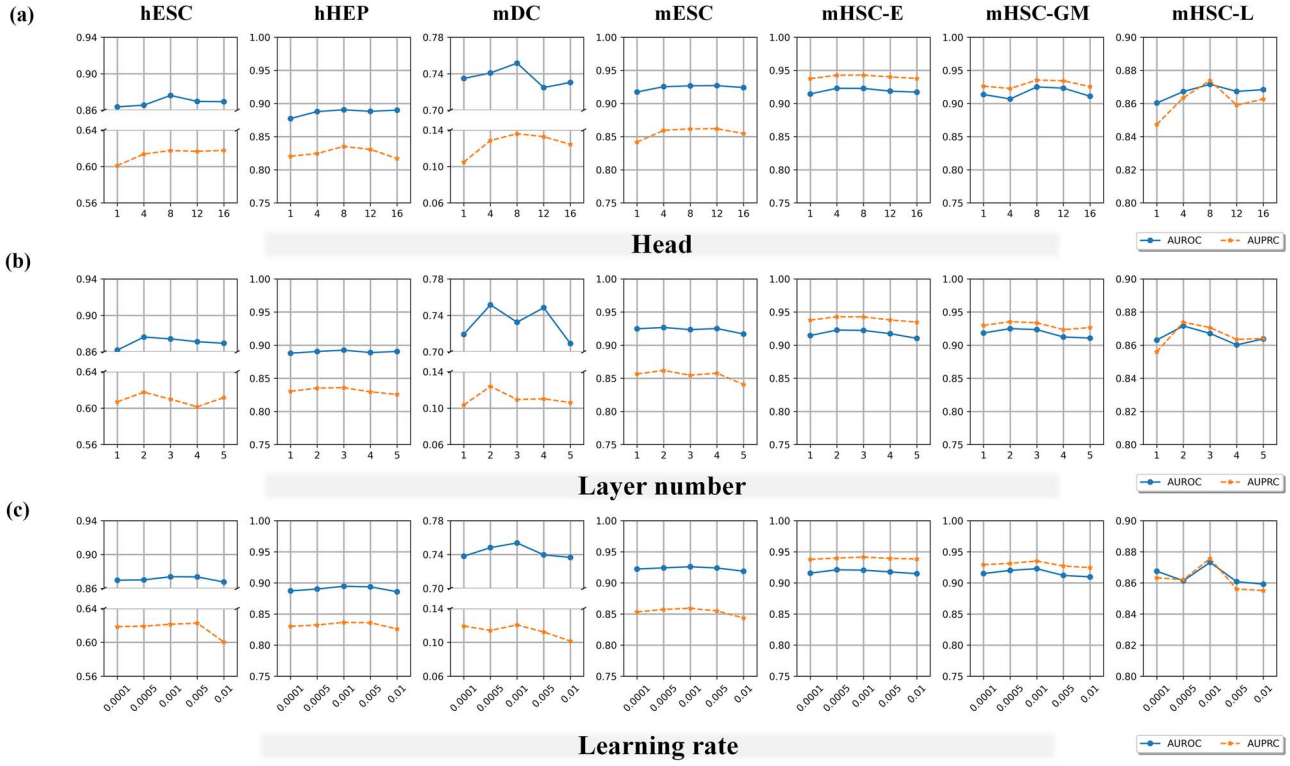


Figure 5. Results of parameter experiments. The impact of (a) TransformerConv attention head number, (b) depth of GNN encoder, and (c) learning rate on model performance.

Effectiveness analysis of single-cell data imputation

Single-cell gene expression data exhibit high sparsity, where numerous zero values do not necessarily indicate true gene silencing. Instead, they often result from technical constraints such as insufficient sequencing depth or limited detection sensitivity, a phenomenon referred to as 'dropout' [42]. These artifacts generate false zeros in the expression matrix, which may distort downstream analyses. In GRN inference, such distortions can lead to the omission of potential regulatory interactions.

Given the high sparsity inherent to scRNA-seq data, we further examined whether expression imputation could enhance the performance of ATFGRN. The raw expression data are preprocessed with DeepImpute [43], a deep learning-based imputation method. DeepImpute employs deep neural networks to estimate and fill in missing values, producing a corrected expression

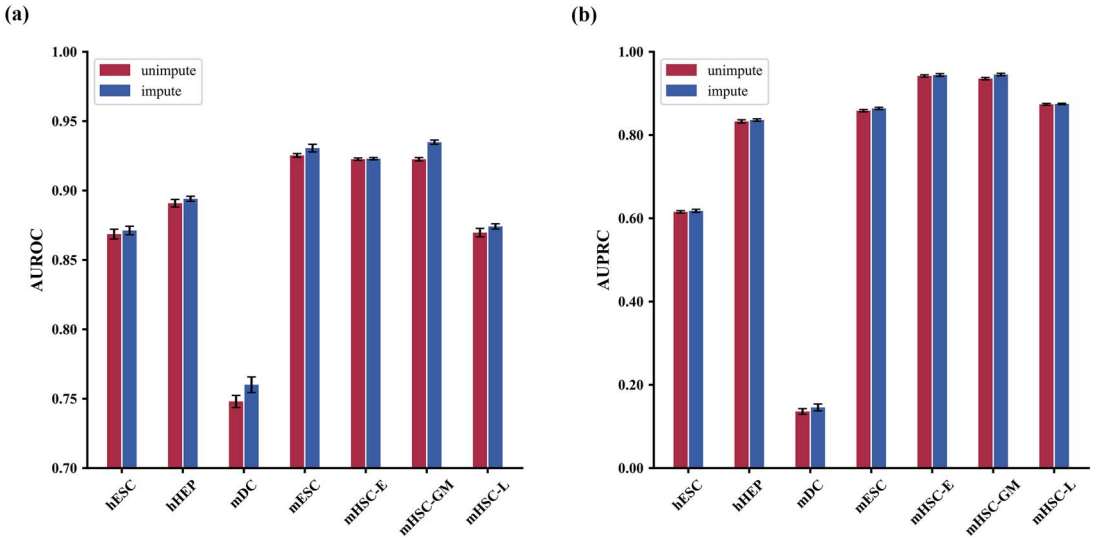


Figure 6. The performance of ATFGRN on imputed and unimputed benchmark datasets with cell-type-specific networks.

matrix. Considering the inherent randomness of deep learning processes, the imputation is repeated 10 times for each dataset, and the aggregated results are summarized in Fig. 6. Detailed numerical statistics appear in Supplementary Tables S5–S6. In addition, we computed the sparsity (i.e. the proportion of zero values) of each dataset before imputation to better contextualize the effects of imputation; these results are summarized in Supplementary Table S7.

The model is trained with both the original and imputed expression data, and performance is evaluated through AUROC and AUPRC metrics across all datasets. The results indicate that performance improvements from imputation primarily appear in datasets that are more constrained by sparsity, contain fewer cells, or possess sparser reference networks. For example, in the mDC dataset (sparsity 21.70%), imputation increased AUROC by 1.61% and AUPRC by 7.11%. The mHSC-GM dataset showed similar gains, with AUROC improving by 1.34%. These findings suggest that, under data-limited conditions, imputation can partially compensate for information loss caused by dropout events.

In contrast, datasets with richer expression information or denser reference networks, such as mHSC-E, exhibited only negligible improvements, with AUROC rising by merely 0.03%. This indicates that when sufficient supervisory information is available, ATFGRN already demonstrates strong robustness to sparsity. Moreover, in cases of extremely high sparsity (e.g. hHEP and hESC) or very low sparsity (e.g. mESC), the effect of imputation remains limited. In highly sparse datasets, severe information loss constrains the recoverable signal, while in low-sparsity datasets, the original expression matrix already provides sufficient information for model training.

Overall, the effectiveness of imputation is jointly influenced by multiple factors, including data sparsity, the number of cells, and the density of the reference network, rather than determined by any single characteristic. These results indicate that ATFGRN performs robustly on unimputed data, and imputation is best regarded as a conditional enhancement strategy-particularly useful for datasets with high sparsity or limited sample size.

Impact of training dataset size on model stability

Most currently constructed GRNs depend heavily on high-throughput experiments or database-based inferences, which provide a limited number of verified regulatory interactions and produce highly sparse network structures. This sparsity restricts the available data for training and makes the model's adaptability to different data scales a key factor in evaluating its effectiveness. To assess ATFGRN's performance under varying data volumes, four representative datasets are selected: the human-derived hESC500 and hHEP500, and the mouse-derived mHSC-GM500 and mHSC-L500. For each cell type-specific network, 10%, 20%, 30%, 40%, and 50% of the target genes associated with each TF are randomly sampled as training subsets. Model performance at these different data scales is evaluated with AUROC and AUPRC metrics.

We conducted 10 independent runs of ATFGRN on each of the four datasets and visualized the distribution of AUROC and AUPRC scores using boxplots, as shown in Fig. 7(a) and (b). The results reveal that model performance consistently improves as the training data size increases, reflecting the supervised approach's dependence on sufficient data volume. When trained with only 10% of the data, ATFGRN performs relatively weakly; however, performance stabilizes as the training proportion reaches ~40%. The model maintains stable outcomes even on smaller datasets, avoiding problems of inadequate gene feature learning. Under larger training conditions, ATFGRN continues to manage complex regulatory structures effectively, which further demonstrates its robustness and adaptability across diverse data scales.

Case study

To evaluate the ability of ATFGRN to predict unknown gene regulatory relationships across different species, two representative cell-specific datasets are selected: the human embryonic stem cell dataset (hESC500) and the mouse embryonic stem cell dataset (mESC500). From each dataset, three TFs with essential biological functions are selected. For each TF, all potential target genes in the corresponding cell line are paired to construct test samples. To prevent information leakage, samples appearing in the training set are removed. The trained model processes these test samples, and for each TF, the top 20 target genes with the highest prediction scores are identified as potential regulatory targets. The prediction outcomes are listed in Tables 2 and 3, and the corresponding regulatory networks are visualized in Fig. 8.

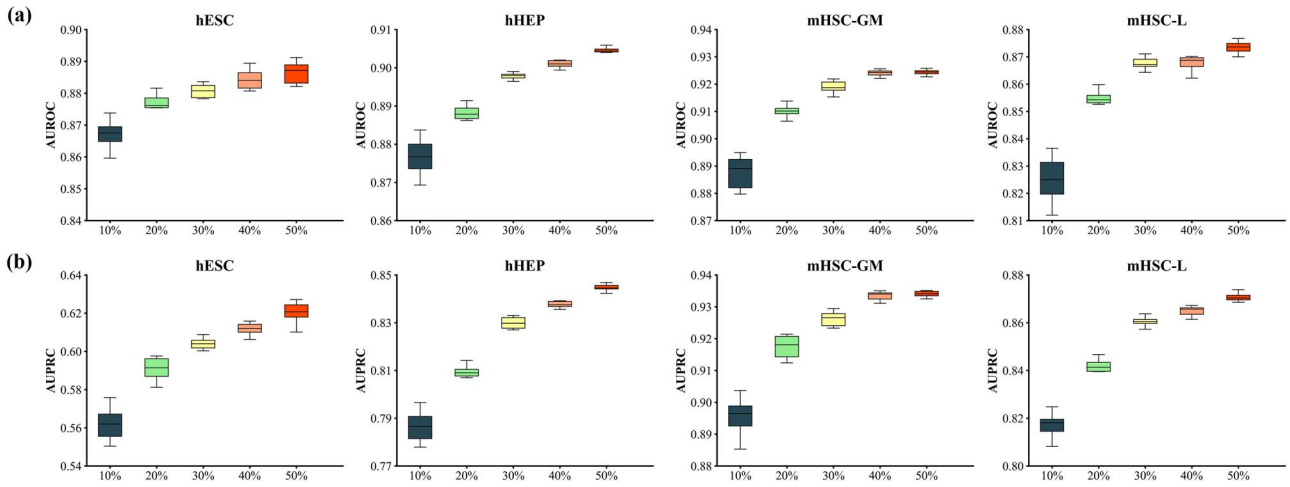


Figure 7. AUROC (a) and AUPRC (b) results of ATFGRN under different training dataset sizes.

Table 2. Top 20 TF–gene pairs from three TFs in the hESC 500 dataset

TF	Target	Evidence	TF	Target	Evidence	TF	Target	Evidence
TFAP2A	APBB2	Ground-truth	NANOG	RBFOX2	Ground-truth	TEAD4	TFAP2A	Ground-truth
TFAP2A	LTBP1	Ground-truth	NANOG	MECOM	Ground-truth	TEAD4	SLC39A8	Ground-truth
TFAP2A	THAP9	Ground-truth	NANOG	APBB2	hTFtarget	TEAD4	ZNF142	Unconfirmed
TFAP2A	PLK2	Ground-truth	NANOG	EGR1	PMID:39044249	TEAD4	TRAPPC8	Unconfirmed
TFAP2A	APC	Ground-truth	NANOG	CD200	Ground-truth	TEAD4	YPEL5	hTFtarget
TFAP2A	HSD17B11	Ground-truth	NANOG	TCF7L1	Ground-truth	TEAD4	NFIA	Ground-truth
TFAP2A	ETS1	Ground-truth	NANOG	MAML3	hTFtarget	TEAD4	TCF7L1	Ground-truth
TFAP2A	BRWD1	Ground-truth	NANOG	SLC39A8	Unconfirmed	TEAD4	INSIG2	Ground-truth
TFAP2A	NR2C1	Ground-truth	NANOG	DTL	Unconfirmed	TEAD4	RPP40	Ground-truth
TFAP2A	CD200	Ground-truth	NANOG	KLF7	Ground-truth	TEAD4	HDAC9	hTFtarget
TFAP2A	FAM117B	Ground-truth	NANOG	HDAC9	Ground-truth	TEAD4	RGPD2	Ground-truth
TFAP2A	NEDD4	Ground-truth	NANOG	AFF4	hTFtarget	TEAD4	NFAT5	Ground-truth
TFAP2A	PRKACB	Ground-truth	NANOG	THAP9	Ground-truth	TEAD4	POU2F1	Unconfirmed
TFAP2A	KLHL24	Ground-truth	NANOG	FAM117B	Ground-truth	TEAD4	HESX1	Ground-truth
TFAP2A	AFF4	Ground-truth	NANOG	SSB	Ground-truth	TEAD4	AFF4	Unconfirmed
TFAP2A	KLHL28	Ground-truth	NANOG	SORL1	hTFtarget	TEAD4	ZFYVE16	Ground-truth
TFAP2A	RPP40	Ground-truth	NANOG	ESRRG	Ground-truth	TEAD4	THY1	Unconfirmed
TFAP2A	ARFGAP3	Ground-truth	NANOG	ANKRD28	Ground-truth	TEAD4	CDC7	Ground-truth
TFAP2A	PDGFA	hTFtarget	NANOG	TRAF6	Unconfirmed	TEAD4	ASXL1	Ground-truth
TFAP2A	SLC36A4	Ground-truth	NANOG	PARP1	PMID:28842672	TEAD4	TRAF6	Ground-truth

To validate the predicted regulatory relationships, we queried human and mouse TF-Target gene pairs using the hTFtarget [44] and Cistrome DB [45], both of which are widely used resources for TF-Target gene pair validation. Results showed that, in the hESC500 dataset, among the 60 predicted TF-Target gene regulatory pairs, 43 were supported by known regulatory networks, and an additional nine were validated through the hTFtarget database and related literature, totaling 52 experimentally supported predictions. In the mESC500 dataset, only three predicted pairs were not validated.

Overall, ATFGRN demonstrates strong capability in identifying unknown gene regulatory relationships across human and mouse cell types. It effectively characterizes complex regulatory interactions between TFs and their target genes, providing meaningful candidates for subsequent biological validation.

Conclusion

This study develops ATFGRN, a GRN inference framework based on a multi-branch fusion architecture. The design integrates single-cell transcriptomic data with prior regulatory knowledge to explore gene regulation from multiple perspectives. ATFGRN consists of three core branches that construct the expression similarity network, combine raw gene expression with regulatory network information, and extract local subgraph topological features. Through weighted fusion of these branches, the framework merges diverse feature types and significantly improves the identification of regulatory edges. Comprehensive evaluation on multiple single-cell datasets and their reference networks shows that ATFGRN surpasses mainstream approaches in AUROC and AUPRC metrics, achieving top performance across most test sets. While the results demonstrate strong performance, there remains room for further improvement and exploration. The complexity of single-cell data and regulatory networks suggests that incorporating additional omics data, such as single-cell ATAC-seq or epigenomics, may support multi-omics integration and enrich input information for more accurate and interpretable inference. In addition, the current negative sample selection depends on a large candidate pool, which may introduce false negatives. Refining this sampling process can reduce such errors and strengthen the framework's precision in identifying true regulatory relationships. Overall, the multi-branch fusion

Table 3. Top 20 TF–gene pairs from three TFs in the mESC 500 dataset

TF	Target	Evidence	TF	Target	Evidence	TF	Target	Evidence
SIN3A	FOXJ3	Ground-truth	HCFC1	ASXL1	Ground-truth	SOX2	SFI1	Ground-truth
SIN3A	SERTAD2	Ground-truth	HCFC1	CNOT8	Ground-truth	SOX2	GABBR1	Ground-truth
SIN3A	MLLT1	Ground-truth	HCFC1	DCP1A	Ground-truth	SOX2	NANOG	Ground-truth
SIN3A	KLF3	Ground-truth	HCFC1	KLF13	Ground-truth	SOX2	POU2F1	Ground-truth
SIN3A	INTS12	Ground-truth	HCFC1	INTS12	Ground-truth	SOX2	POU5F1	Ground-truth
SIN3A	DNMT3A	Ground-truth	HCFC1	MME	Ground-truth	SOX2	NRF1	Ground-truth
SIN3A	NOTCH2	Ground-truth	HCFC1	GSR	Cistrome DB	SOX2	HIST1H2BF	Ground-truth
SIN3A	ZFHX3	Ground-truth	HCFC1	EIF4A2	Ground-truth	SOX2	TCF12	Ground-truth
SIN3A	GLUL	Ground-truth	HCFC1	ATF7	Ground-truth	SOX2	HIST1H2AC	Ground-truth
SIN3A	FUBP3	Ground-truth	HCFC1	SND1	Ground-truth	SOX2	GLS	Ground-truth
SIN3A	CNOT2	Ground-truth	HCFC1	GLUL	Ground-truth	SOX2	CNOT2	Unconfirmed
SIN3A	SND1	Ground-truth	HCFC1	GABBR1	Cistrome DB	SOX2	MED12	Ground-truth
SIN3A	RAD54B	Ground-truth	HCFC1	HIST1H2BN	Ground-truth	SOX2	SOCS2	Ground-truth
SIN3A	TMF1	Ground-truth	HCFC1	ODC1	Ground-truth	SOX2	ING1	Ground-truth
SIN3A	PMS1	Ground-truth	HCFC1	HIST1H2AC	Ground-truth	SOX2	SMARCD1	Ground-truth
SIN3A	POLR2J	Ground-truth	HCFC1	COPS2	Ground-truth	SOX2	LDB1	Ground-truth
SIN3A	CTNNB1	Ground-truth	HCFC1	HIST1H4K	Ground-truth	SOX2	NFE2L1	Unconfirmed
SIN3A	PDIA6	Ground-truth	HCFC1	RBM14	Ground-truth	SOX2	BRCA2	Cistrome DB
SIN3A	HMGXB4	Ground-truth	HCFC1	NOTCH2	Ground-truth	SOX2	DCP1A	Cistrome DB
SIN3A	HIST1H3H	Ground-truth	HCFC1	HIST1H2AF	Unconfirmed	SOX2	CALU	Cistrome DB

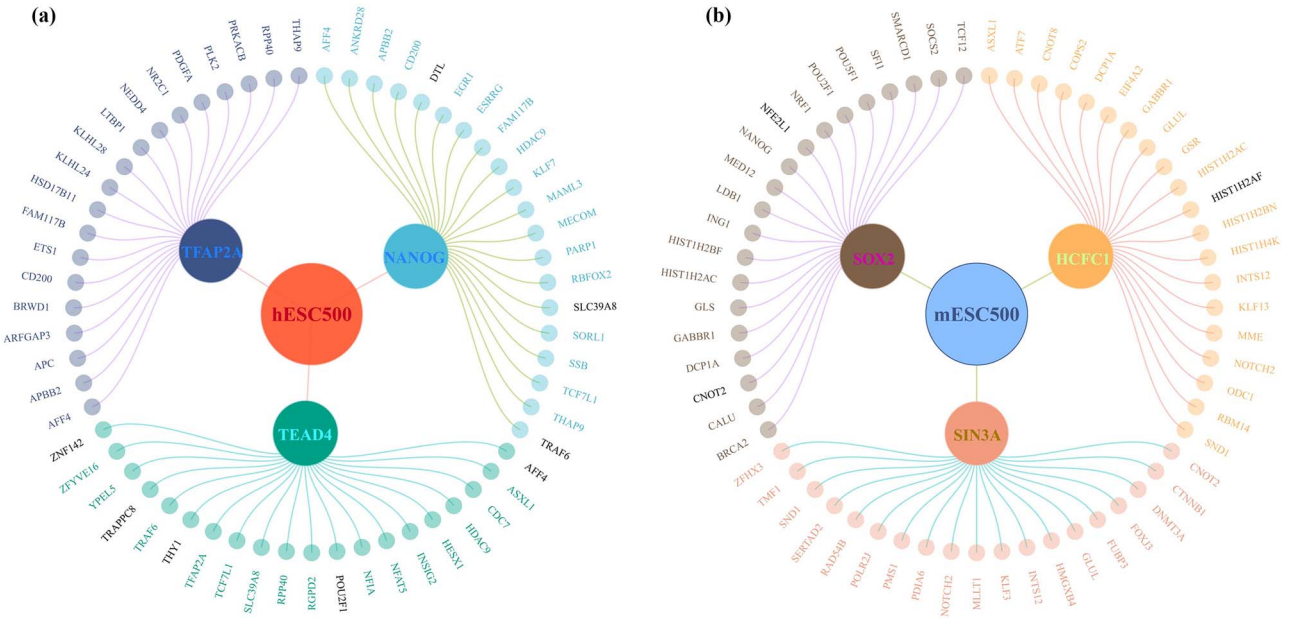


Figure 8. (a) Top 20 TF-gene pairs from three TFs in the hESC 500 dataset, (b) top 20 TF-gene pairs from three TFs in the mESC 500 dataset. Black font indicates unconfirmed TF-gene pairs.

design of ATFGRN effectively integrates expression-based and structural information, demonstrating stable performance and high predictive accuracy in GRN inference. The framework provides a reliable foundation for advancing the analysis and understanding of gene regulatory mechanisms.

Key Points

- ATFGRN is a multi-perspective graph-based framework that accurately infers gene regulatory networks from single-cell transcriptomic data.
- It performs topology-feature fusion by integrating information from local subgraph structures, expression-network interactions, and expression-based similarity through attention-guided combination.
- Experiments on multiple single-cell transcriptomic datasets demonstrate consistent performance gains over existing approaches, verifying the robustness and broad applicability of the framework.

Supplementary data

Supplementary data are available at *Briefings in Bioinformatics* online.

Conflict of interest: The authors assert that they have no conflicts of interest to this study.

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Data availability

The datasets, codes and corresponding results are available at https://github.com/Evert-zrt/ATFGRN.

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Problem Solving Protocol